

14: GENCHEM2 LAB SPRING 2023 2



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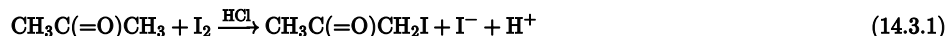
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14.3: Acetone Iodine kinetics

Introduction

In this lab, we will apply the concepts of reaction rate, order of reaction and catalyst. Our example reaction is the iodination of acetone catalyzed by hydrochloric acid.



The reaction is zero order in iodine. You will figure out the order in acetone and the order in HCl, and the rate constant. Then, you will predict which initial concentrations are necessary to make the reaction run for exactly 2 minutes (120 s).

We will figure out the reaction rate by visual observation of the iodine. Iodine in water is brownish-yellow, and all other species are colorless. In the experiments, you will start a timer once iodine is added to the other substances in the reaction mixture, and measure the time until the solution appear colorless. We will run the reaction at room temperature. Try not to warm up any of the reactants or the reaction mixture with your hands or other sources of heat (e.g. sunlight near window).

Experimental design

Finding the rate

Usually, the reaction rate of a reaction changes over time as reactants are depleted. Also, the concentrations change over time. To be able to "plug in" the rate and the concentrations into the rate law and determine the order of reactions, the typical strategy would be to measure initial rate.

Today, however, we are measuring an average rate. We can get away with that because it turns out the rate stays constant over the course of the experiment, which is unusual. There are three reasons for this:

1. The HCl concentration does not change (catalyst)
2. The acetone concentration does not change significantly (in large excess over iodine)
3. The iodine concentration changes a lot, but it does not affect the rate (zero order in iodine)

To calculate the rate, we consider the time it took for the iodine to "fade", and the initial concentration of the iodine. For example, if the initial iodine concentration is 0.001 mol /L, and it takes 180 s for the reaction mixture to become clear, the rate would be:

$$\text{rate} = \frac{\Delta[\text{I}_2]}{\Delta t} = \frac{1.0 \times 10^{-2}\text{M}}{180\text{s}} = 5.5 \times 10^{-5}\text{M}^{-1} \text{ s}^{-1} \quad (14.3.2)$$

Varying concentrations to find the order

You will set up multiple reactions, varying the concentrations in the reaction mix. To quickly figure out the concentrations, it makes sense to always keep the final volume the same (25 mL), and use the dilution law to figure out concentrations (or volumes you need to add to get the desired concentration). Here is the dilution law:

$$c_{\text{mix}} V_{\text{mix}} = c_{\text{stock}} V_{\text{stock}} \quad (14.3.3)$$

If you want to know the (initial) concentration of a component in the reaction mixture, plug in the volume and concentration of the stock solution. For example, if our 25 mL reaction mixture contains 5 mL iodine stock with an original concentration of 0.005 M, we get:

$$c_{\text{mix}} = c_{\text{stock}} \frac{V_{\text{stock}}}{V_{\text{mix}}} = 0.005\text{M} \frac{5\text{mL}}{25\text{mL}} \quad (14.3.4)$$

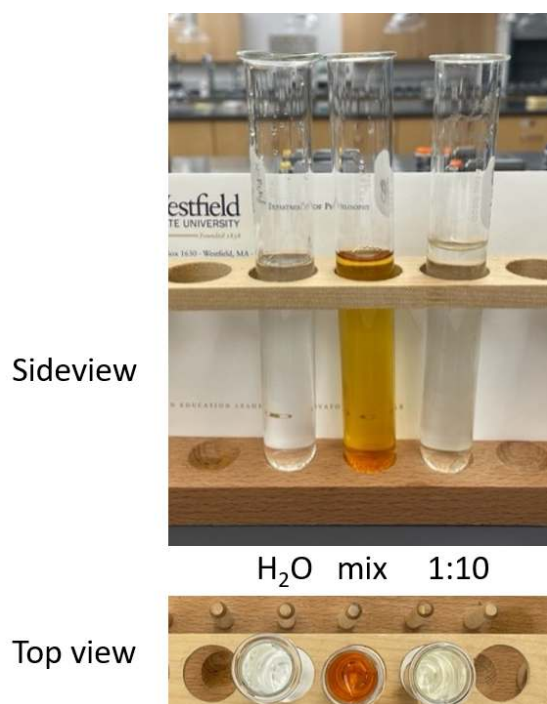
If you want to know how much of the 4-M acetone stock solution you need to get 0.8 M in the reaction mixture, you solve for the volume of stock solution:

$$V_{\text{stock}} = V_{\text{mix}} \frac{c_{\text{mix}}}{c_{\text{stock}}} = 25\text{mL} \frac{0.8\text{M}}{4\text{M}} \quad (14.3.5)$$

Procedure

This is the procedure for measuring a single rate. You will run this procedure twice for each different reaction mixture. You can figure out the concentrations of the mixture from the recipe (the volumes mixed) using equation 4 or [this spreadsheet](#). Make sure the volumes in your recipe always add up to 25 mL.

1. Mix water, acetone stock and hydrochloric acid stock in a 125 mL Erlenmeyer. The reaction can not proceed at this stage (no iodine).
2. At your bench, add the correct amount of iodine stock (5 mL) to the mixture and start the timer.
3. For better visibility of the color change, pour the reaction mixture into a medium test tube, and observe from the top, with a white paper under the test tubes. Prepare a large test tube with 25 mL water and one with 0.5 mL iodine solution in 24.5 mL of water for comparison (see picture below: the iodine solution labeled 1:10 looks like water in the side view, but has a faint brownish tint in the top view).
4. Stop the timer once the solution is colorless. Record the time in your lab notebook and calculate the rate for your experiment (like in equation 2).
5. Keep the test tube with water and the 1:10 mix for the remainder of the runs. Discard the reaction mix, rinse the 125 mL Erlenmeyer and the large test tube and invert to dry so you can use them again for the next run



Determining the orders of reaction

The easiest way to find an order of reaction is to halve the concentration of the species you are interested in while keeping all other concentrations the same. If the reaction time stays the same, it is zero order in the species whose concentration you changed. If the reaction time doubles (the rate halves), it is first order. If the change is more extreme, it is a higher order than one. If you learned a mathematical way of determining the reaction order in your lecture section, feel free to use that method. Because of experimental errors, we don't expect the reaction order to come out exactly as 0, 1, or 2, and you should feel free to round your result as you formulate the rate law.

To find out two orders of reaction (with respect to acetone and with respect to hydrochloric acid), you need three sets of concentrations (see [data sheet](#)). You will run each set in duplicate, so you end up running a total of six experiments.

We have aqueous stock solutions containing acetone ($c = 4 \text{ M}$), hydrochloric acid ($c = 1 \text{ M}$), and iodine ($c = 0.005 \text{ M}$). For the first set of concentrations, you will use 10 mL acetone solution, 5 mL water, and 5 mL hydrochloric acid (which is corrosive and should

be handled under the hood until diluted), giving a subtotal of 20 mL. Measure out 5 mL iodine solution in a graduated cylinder, and start the reaction at your bench by adding the iodine to the other components in the Erlenmeyer. Record the time and repeat the experiment once to check for consistency.

For the second set of concentrations, we are using half the volume of acetone, and for the third set of concentrations, we are using half the volume of hydrochloric acid (compared to the first set). The volumes are given in the data sheet. Calculate the volume of water you need. You should check your recipe before you make the reaction mixture. Make sure the volumes in your recipe always add up to 25 mL.

Predicting the rate of reaction

Once you know the form of the rate law (i.e. all the orders of reaction), you can plug in all the concentrations and the measured rate to calculate the rate constant. It should be the same for all of your trials (if the temperature is constant), so you can average it to average out random errors. Once you have the rate constant, you can predict the rate (and the reaction time) for any set of concentrations. We will have a contest to see who can get their reaction to end after exactly 2 minutes (120 s), or closest to that time.

The contest will be done bench by bench. Once everyone has calculated the concentrations they want to use, and has prepared their reaction mixture (minus the iodine), you can ask for the contest to start. Your instructor starts a countdown, and you will add the iodine to the flask when the timer starts. Then, you will transfer your reaction mixture to a large test tube and place it in a test tube rack in the front. A doc camera will project the image of the tubes on the screen, so everyone in the room can see. Whenever a test tube turns colorless, we will record the time. The reaction completing closest to 120 s wins our appreciation for good calculations and good experimental technique (with a bit of luck).

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14.13: Freezing Point Depression

Goals

- Measure freezing point depression.
- Determine the Van't Hoff factor for two different solutes

Safety

- The substances we are using today are household chemicals (table salt and sugar). However, the usual lab precautions apply.

Equipment and materials

- From your drawer: 50 mL beaker, 100 mL beaker
- On your bench: temperature probe
- At the balances: sucrose (s), sodium chloride (s)
- For task 4: sieve, calorimeter
- At a station: ice

Background

A colligative property is when a solute affects a solution property like its freezing or boiling point, and the effect depends only on the amount of particles dissolved, no matter which substance you use. In this lab we will investigate the effect of the different solutes on the freezing point of aqueous solutions.

You should observe that as the solute concentration increases the freezing point $T_{fp}(\text{solution})$ decreases (the freezing point depression ΔT increases), and this can be described by the following equation:

$$\Delta T = i \cdot k_f \cdot \frac{n_{\text{solute}}}{m_{\text{solvent}}} \quad (14.13.1)$$

where

i = Van't Hoff Factor

k_f = freezing point point constant

n = amount

m = mass

and

$$\Delta T = T_{fp}(\text{pure solvent}) - T_{fp}(\text{solution})$$

The value of k_f for the solvent water is 1860 °C g/mol, or 1.86 °C kg/mol. Check your units to decide which format is more convenient in this case.

Experimental Design Considerations

We are making the assumption that at the instance the ice melts the mixture of ice and water is at the freezing point. There can be [multiple reasons](#) why this is wrong, and we will discuss this in our pre-lab discussion.

Experimental Procedures

Record all observations in your lab notebook, and summarize your calculations and interpretations in [this worksheet](#).

Part 1: Freezing point determination, pure water

1. Obtain a temperature probe and turn it on. Measure the mass of a clean, dry 100 mL beaker.
2. Pour approximately 25 mL of DI water into the 100 mL beaker
3. Place temperature probe into pure water in beaker
4. Add around 20 mL of fresh crushed ice to a 50 mL beaker.
5. Quickly add the ice to the 100 mL beaker, gently swirl the temperature probe. [Record](#) the lowest temperature, and the temperature just as the last bit of ice melts.

6. Weigh the mass of the beaker with the water and melted ice (so now just water). Record to the precision of the instrument. Clean and dry the 100 mL beaker and record the empty mass to the precision of the instrument.
7. Discard the water in the sink.
8. In your further calculations, use the lowest recorded temperature as the freezing point of water. The temperature probes are not calibrated, but are very good at measuring temperature differences. Don't be worried if the reading on your temperature probe is different from that of your neighbor, and don't swap temperature probes.

Part 2: Freezing point determination, sucrose solution

You can do part 2 and part 3 in any order, depending which balance is available.

1. Weigh approx 0.05 mol of sucrose (molar mass is 342.3 g/mol), record the exact mass, and transfer to the clean and dry 100 mL beaker.
2. Add approximately 25 mL water.
3. Place temperature probe into the solution in the beaker and swirl and stir to dissolve the sucrose completely.
4. Add around 20 mL of fresh crushed ice to the 50 mL beaker.
5. Quickly add the ice to the 100 mL beaker, gently swirl the temperature probe. Record the lowest temperature, and the temperature just as the last bit of ice melts.
6. Weigh the mass of the beaker with the sucrose solution and melted ice (so now a diluted sucrose solution). Record to the precision of the instrument. Clean and dry the 100 mL beaker and record the empty mass to the precision of the instrument. Then, calculate the mass of water (solvent) from the available data.
7. Calculate the van't Hoff factor for sucrose and record your work in the lab summary.

Part 3: Freezing point determination, sodium chloride solution

1. Weigh approx 0.05 mol of sodium chloride (molar mass is 58.44 g/mol), record the exact mass, and transfer to the clean and dry 100 mL beaker.
2. Add approximately 25 mL water.
3. Place temperature probe into the solution in the beaker and swirl and stir to dissolve the sodium chloride completely.
4. Add around 20 mL of fresh crushed ice to the 50 mL beaker.
5. Quickly add the ice to the 100 mL beaker, gently swirl the temperature probe. Record the lowest temperature, and the temperature just as the last bit of ice melts.
6. Weigh the mass of the beaker with the sodium chloride solution and melted ice (so now a diluted sodium chloride solution). Record to the precision of the instrument. Clean and dry the 100 mL beaker and record the empty mass to the precision of the instrument. Then, calculate the mass of water (solvent) from the available data.
7. Calculate the van't Hoff factor for sodium chloride and record your work in the lab summary.

Part 4: Try a different experimental strategy

With a partner, measure the freezing point of a solution of sodium chloride again, using a different experimental strategy based on the calorimeters we built last semester. This method is a bit faster but requires more equipment (work with a calorimeter instead of a beaker to minimize heat transfer, or use the minimum temperature in the calculation, quickly removing the remaining ice by pouring the slush through a sieve). As you are going through the steps, reflect on the pros and cons of this and the original method.

1. Weigh approx 0.05 mol of sodium chloride (molar mass is 58.44 g/mol), record the exact mass, and transfer to calorimeter cup.
2. Add approximately 25 mL water from a 100 mL beaker.
3. Place temperature probe into the solution in the beaker and swirl and stir to dissolve the sodium chloride completely. Place the calorimeter cup into the calorimeter.
4. Add around 20 mL of fresh crushed ice to the 50 mL beaker.
5. Quickly add the ice to the calorimeter cup, and gently stir with the temperature probe. Once the temperature stops decreasing (make sure there is still ice present), record the temperature, remove the temperature probe, and pour the mixture through a sieve into a dry 100 mL beaker. Discard the ice in the sieve.
6. Weigh the mass of the beaker with the sodium chloride solution. Record to the precision of the instrument. Clean and dry the 100 mL beaker and record the empty mass to the precision of the instrument. Then, calculate the mass of water (solvent) from the available data.

7. Calculate the van't Hoff factor for sodium chloride and record your work in the lab summary.

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14.15: Paper chromatography

Objectives

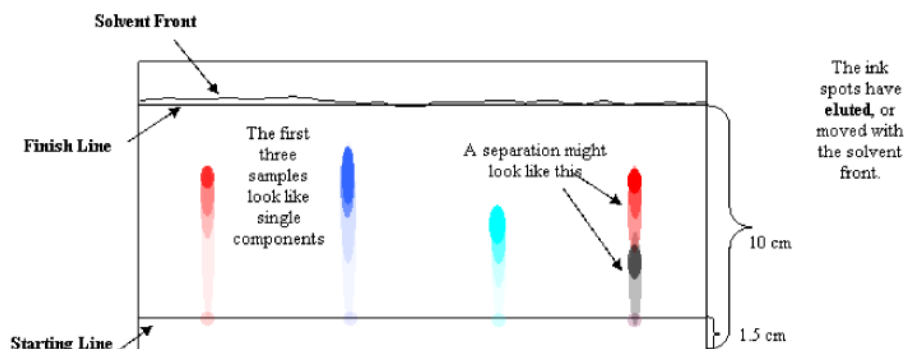
- To use paper chromatography to identify whether certain food dyes are pure substances or mixtures.
- To obtain a paper chromatogram of various food dyes
- To identify components of food dyes by calculating R_f values

Chromatography is a method of physically separating *mixtures* into its individual components. It is a common laboratory technique used to identify unknown components in mixtures. The "Chroma" in chromatography refers to colors. While early chromatography (and our experiment today) relied on our eyesight to read a chromatography result, modern instrumentation has other detection methods (absorbance in the ultraviolet or infrared, flame ionisation, mass spectrometry and many other techniques, depending on the type of samples you are separating).

There are several types of chromatography; all types employ a *mobile phase* or *eluent* (it can be liquid or gas), which is forced through a *stationary phase* (a solid or semi-solid). Mixtures are separated because some components will be more attracted to the stationary phase (and stick to it) while some components will be more attracted to the mobile phase (and travel with it).

By eye, we cannot know if each dye color is a mixture or pure substance. Using chromatography, the components in a sample will migrate along the filter paper at different rates such that they become spread out and separated from each other. The mobile phase takes advantage of differing solubility or polarity of the components in order to separate them. This component traveling process is called *elution*. Chromatography can be used to separate inks, dyes, pigments in plants, or used to determine the chemical composition of many substances.

Figure 1 shows a typical paper chromatogram. There are a few difficulties commonly encountered in the elution process. One problem is that spots tend to spread out as they elute, and can bleed into each other as they proceed up the paper. This can be confusing when interpreting the chromatogram. To avoid this problem, space the spots of sample far apart and make repeated, tiny applications of sample to prevent spreading. Another problem is an uneven solvent front. This can happen if the beaker is nudged – if the mobile phase sloshes inside, the elution trails may travel diagonally, which makes interpretation very difficult.



A component with a given solubility travels along with the mobile phase at one rate, regardless of what other components are present in the sample. If the red part of purple dye travels at the same rate as pure red dye, and both stop in the same place, the two should be the same red dye. The two red spots should have the same *Retention Factor*, R_f . The R_f is the distance, D , traveled by the spot divided by the distance traveled by the eluting solution, or *Solvent Front*, F .

$$R_f = \frac{D}{F} \quad (14.15.1)$$

Comparing the R_f values allows the confirmation of a component in multiple samples because unique components have unique R_f values (for the same solvent mixture at the same temperature).

Procedure

Materials and Equipment

Materials: chromatography paper, candy, and eluting solution.

Equipment: 600-mL beaker, pencil, ruler, binder clip and Whatman #1.

Safety

Be sure to handle only the dry part or the chromatogram when removing it from the beaker. Wash hands thoroughly if the eluting solution touches your skin. Personal protective equipment (PPE) required: closed-toe shoes

Part A: Preparation of dye solutions

1. You will use two pieces of candy, one with a primary color (red, yellow, blue), and one other one (green, orange, brown). You can choose what type of candy. Add two drops of water into a well of the spot plate. Add a piece of candy such that one side of it is suspended in water. After most of the dye is dissolved, turn the candy around to dissolve the dye on the other side. Once most of the dye is dissolved (i.e. you can see the white coating), remove the candy and discard. Do not eat the candy.
2. If you want to have a higher concentration of dye in solution, work together with someone who has the same color, and dissolve the dye from another piece of candy in the same solution. There is plenty of volume for multiple runs, so you can share the resulting dye solution.
3. Repeat for the different colors assigned to you.

Part B: Preparation of Chromatography Paper

1. Wash your hands thoroughly to remove excess oils from your skin. Obtain a ruler and a piece of chromatography paper. Handle the paper only on the edges to avoid leaving fingerprints, as these may hinder the elution process.
2. Place the chromatography paper on a sheet of clean notebook paper or paper towel to avoid picking up dirt or contaminants from the bench top. Orient the paper into a “portrait” position and write your name on the top edge of the paper in one corner. Using a pencil and ruler to measure accurately, draw a straight line across the paper, about 1.5 cm above the bottom edge. This is the **starting line**.
3. Lightly draw two small “X” centered on the starting line so that there is plenty of space between the spots and between spots and edge of the paper.
4. In the center of each X, apply a small amount of dye with a capillary tube (using the solutions you made in part A, see Figure 2 right).



Figure 2: Using the binder clip (left) and applying the sample (right)

5. Attach the top of the paper to a binder clip and insert a pencil through the metal loops (see left panel). Then, lay the pencil across a 600 mL beaker (avoiding the spout) and adjust the paper so it almost touches the bottom.

Part C: Acquisition of Chromatogram

1. Remove the paper strip, clip and pencil assembly from the beaker.
2. Find your eluting solution. Depending which bench you are working at (near the safety shower, near the fire alarm, or middle bench), you will use a different eluting solution. Pour about 25 mL of eluting solution into the beaker.

Bench	Eluting solution
near the safety shower	0.1 % aqueous sodium chloride solution
middle bench	0.1 % aqueous sodium chloride solution containing 5% ethanol
near the fire alarm light	1 % aqueous sodium chloride solution

3. Place a wick (a larger filter paper) into the beaker so it touches the side, and angle the beaker so that the wick gets wet (without pouring out the solution). Get a cover ready (aluminum foil or saran wrap).
4. Gently place the pencil on the rim of the beaker so that the paper is submerged in solution, but the dye spots are not submerged. You should see the solvent move up the paper strip. If necessary, adjust the height. Remember that the spots must be above the liquid level for the experiment to work.
5. Start a timer.
6. Cover the beaker with aluminum foil or saran wrap. Let the solvent move up least 6 cm (two-thirds of the length of the paper strip). This will take some patience.
7. When the solvent front moved far enough, lift the paper out of the solution by lifting the pencil. Let excess eluent drip into the beaker. Mark the position of the solvent front with a pencil (if you forget this step, you can later see the solvent front nicely under UV light). Let the paper air dry, attaching it to a snorkel hood to speed things up. Let it dry completely. Again, this might take some patience.

Part D: Interpretation of Chromatogram

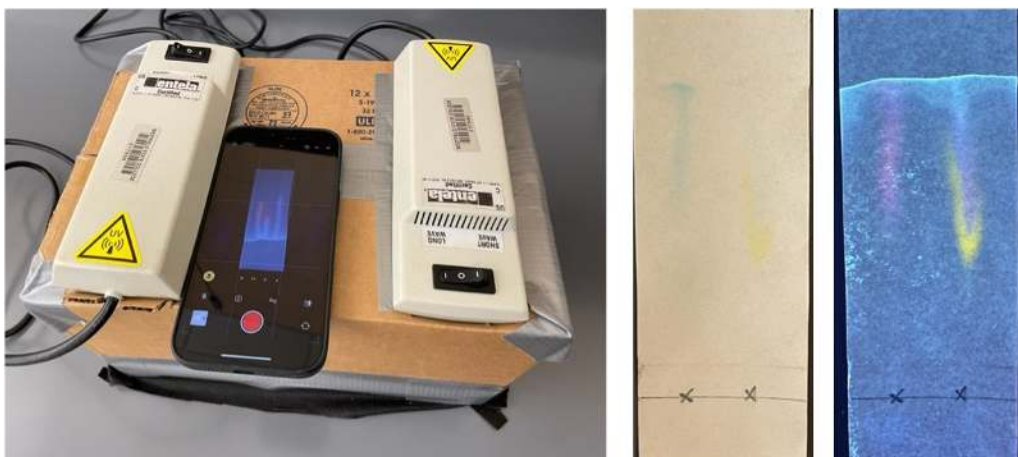


Figure 3: UV illuminator (left) and chromatogram viewed under visible vs. UV light (right)

1. Circle around each color spot. For mixtures, circle each separated color as best as you can.
2. Use the UV-cardboard illuminator (Figure 3 left panel) to see which spots fluoresce in a different color than the visible color. This will also help detecting colors like blue and yellow that might be difficult to see in visible light (Figure 3 right), and visualizing the solvent front after it dried.
3. Use a ruler and draw a plus sign in the center of each spot. Remember, if you separated a mixture, each single color gets a plus sign. Measure the distance from the starting line to each plus sign. Record this distance for each spot on your lab report. These

are the D values, in cm.

4. Measure the distance between the starting line and the solvent front reached. Record this distance. This is the F value, in cm.
Unless your solvent ran unevenly, this should be the same for all samples on a single strip.
5. Calculate the retention factor (R_f) for each spot and record the values in your lab report.
6. Compare the retention factor of samples with multiple spots (dye mixtures) to those with a single spot (pure dyes). What are colors used for the mixtures? Is that what you expected from your experience growing up?

[Lab1 result sheet.pdf](#)

Acknowledgements

Thanks to the Spring 2023 Monday lab section at Westfield State for many suggestions, corrections and the samples pictured here.

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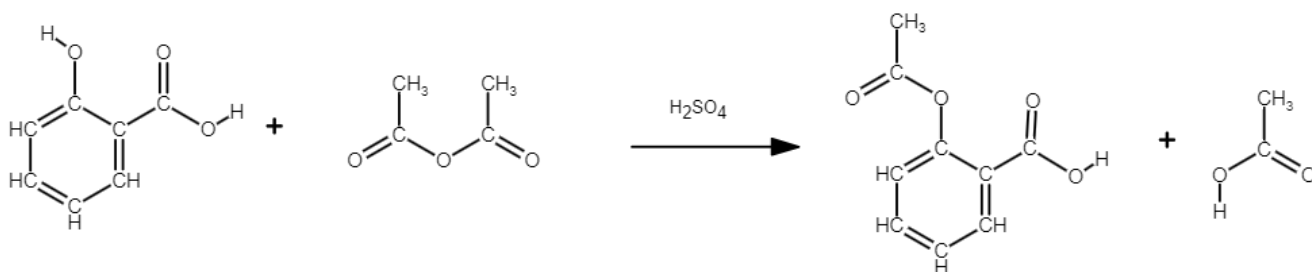
- **2: Paper Chromatography of Gel Ink Pens (Experiment)** by [Santa Monica College](#) is licensed [CC BY-NC 4.0](#).

14.16: Synthesis and Recrystallization of Aspirin

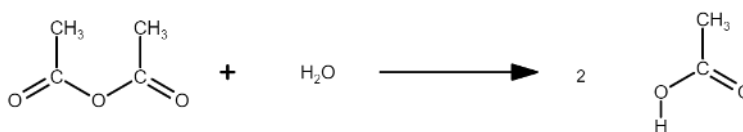
Over history, many compounds obtained from nature have been used to cure ills or to produce an effect in humans. These [natural products](#) have been obtained from plants, minerals, and animals. In addition, various transformations of these and other compounds have led to even more medically useful compounds.

Analgesics are compounds used to reduce pain, antipyretics are compounds used to reduce fever. One popular drug that does both is aspirin. The Merck Index, which is an encyclopedia of chemicals, drugs and biologicals, lists the following information under aspirin: acetylsalicylic acid; monoclinic tablets or needle-like crystals; mp 135 °C (rapid heating); is odorless, but in moist air it is gradually hydrolyzed into salicylic and acetic acids; one gram dissolves in 300 mL of water at 25 °C, in 100 mL of water at 37 °C, in 5 mL alcohol, in 17 mL chloroform.

Synthesis of aspirin (acetylsalicylic acid)



Reaction scheme 1: Salicylic acid and acetic acid anhydride react to form aspirin and acetic acid, catalyzed by sulfuric acid



Reaction scheme 2: Acetic anhydride reacts with water to form two molecules of acetic acid

Week 1: Aspirin Synthesis

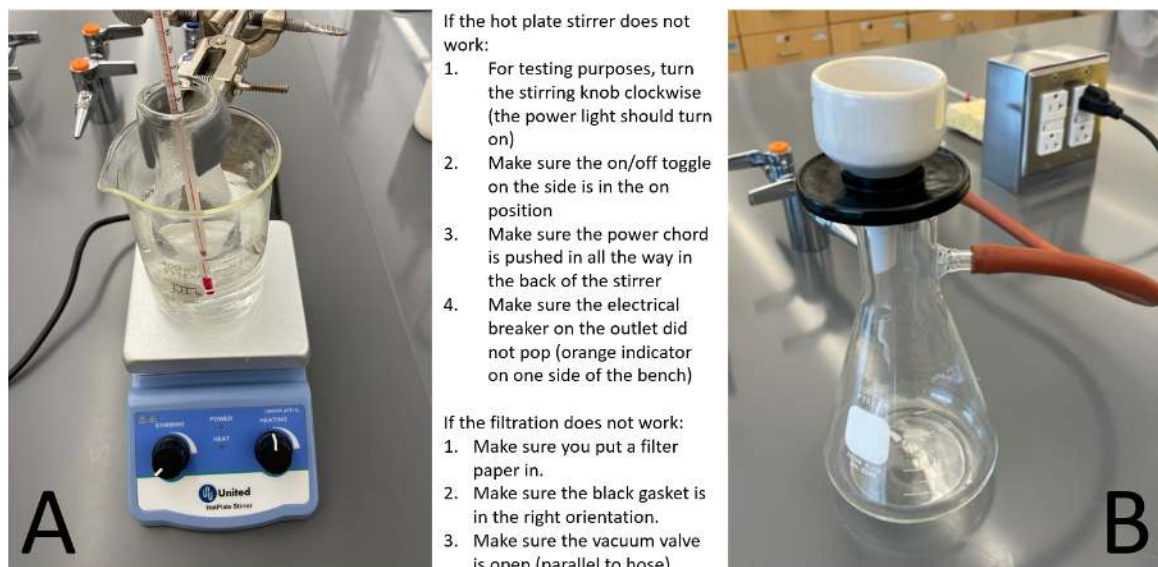
For the first week of this two-week experiment, you will synthesize and recrystallize aspirin.

Synthesis of crude aspirin

1. Set up a hot water bath (400 mL beaker filled to 100 mL with water) on a hotplate and adjust the water temperature to about 75 °C (see photo below - after the water reaches the desired temperature, set the heating dial to the minimal value as in the photo, or turn it off if the temperature is too high).
2. Place 2.0 g (about 15 mmol) of salicylic acid in a 125-mL Erlenmeyer flask.
3. Under the hood, add 5 mL (5.4 g, about 50 mmol) of acetic anhydride, followed by 5 drops of conc. H₂SO₄ (*use the provided dropper, H₂SO₄ is highly corrosive*) and swirl the flask gently most of the salicylic acid dissolves (the remainder will dissolve as you heat the reaction).
4. Heat the flask gently in the hot water bath for 15 minutes to run reaction scheme 1 shown above (if you overheat, you get side products; if you don't heat, you get less of the desired product). The flask will float on the hot water at an angle, which is fine.
5. Allow the flask to cool until you can touch the bottom. Under the hood, add 10 mL of water to react with the excess of acetic anhydride (see reaction scheme below). Bring the flask back to your bench.
6. If acetylsalicylic acid does not begin to crystallize out, scratch the inside walls of the flask with a glass rod or add a seed crystal. Cool the mixture slightly in an ice bath until crystallization is completed. **If the mixture appears milky or there is a whitish**

oil on the bottom, consult your instructor. The product will appear as a solid mass when crystallization is completed. Then, add 50 mL of water and cool the mixture in an ice bath. Do not add the water until crystal formation is complete.

- Vacuum filter the product using wetted round filter paper in a Buchner funnel (see photo below for setup). The total amount will not fit in the funnel. Swirl before you pour to get the chunks transferred, and use the glass rod with rubber policeman to transfer any remaining solids.
- Rinse the crystals several times with small portions (5 mL) of cold water and air dry the crystals on a Buchner funnel by suction until the crystals appear to be free of solvent (no more liquid dripping from the funnel. Place the crude solid on a pre-weighed watch glass and record the mass of the solid. Place a small amount on a watch glass to dry. We will use a portion for the ferric chloride test, and a small amount to measure the melting point next week.



Panel A: Water bath setup (instead of clamping the Erlenmeyer, you can also carefully place it without clamping). Panel B: filtration setup. For troubleshooting tips, see center.

Recrystallization of aspirin

- Place the remaining crude product into a clean 125 mL Erlenmeyer flask. Add ethanol (about 1 mL for every gram of crude product) and warm in the water bath at about 50°C. If lots of solid remains, add more ethanol in 1-mL portions. Once most or all is dissolved, add 40 mL water. Cool down to room temperature and make sure crystals form (you can use the scratching method or add a tiny crystal of crude product if crystals don't form on their own) before you cool down further in an ice bath. **If you get a milky liquid instead of solid crystals, consult your instructor.**
- Vacuum filter the product using a Buchner funnel.
- Rinse the crystals several times with small portions (5 mL) of cold water and air dry the crystals on a Buchner funnel by suction until the crystals appear to be free of solvent. Test this recrystallized product for the presence of unreacted salicylic acid using the ferric chloride test (procedure below) and keep some for measuring the melting point next week.
- Fill out the [result sheet](#).
- Record the mass and calculate the relative yield in percent.
- Follow your instructor's directions to properly store your product to dry by next week.

Week 2: Aspirin Characterization

For the second week of this experiment, you will perform the following tests after your aspirin product has completely dried.

Ferric chloride test for salicylic acid

Phenols are compounds that have a hydroxyl (-OH) group attached directly to a benzene ring. Salicylic acid is a phenol while aspirin is not. Formation of an iron-phenol complex with Fe(III) gives a definite color ranging from red to violet, depending upon the particular phenol present.

1. Add 10 drops of aqueous 1% ferric chloride solution to a test tube containing a few crystals of the compound to be tested dissolved in 5 mL water and note the color.
2. Do this test with (a) salicylic acid, (b) your crude product and (c) your recrystallized product.

Melting point determination

This is done in batches of three samples (crude product, recrystallized product, store-bought product). The idea is to start with low temperature, and watch the sample visually as you ramp up the temperature, writing down the temperature at which each sample melts:

1. Make sure the temperature of the melting point apparatus is below 70 °C.
2. Place your samples in the melting point apparatus. Make sure you can see the solids.
3. Turn on the heater. Watch the samples and the temperature (two people).
4. Once the last sample is melted, turn off the heater.
5. **Make sure the heater is turned off** to allow the next group of measurements to proceed as soon as possible.

HPLC analysis

Follow your instructor's directions to analyze your aspirin product using high performance liquid chromatography (HPLC).

Contributors and Attributions

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